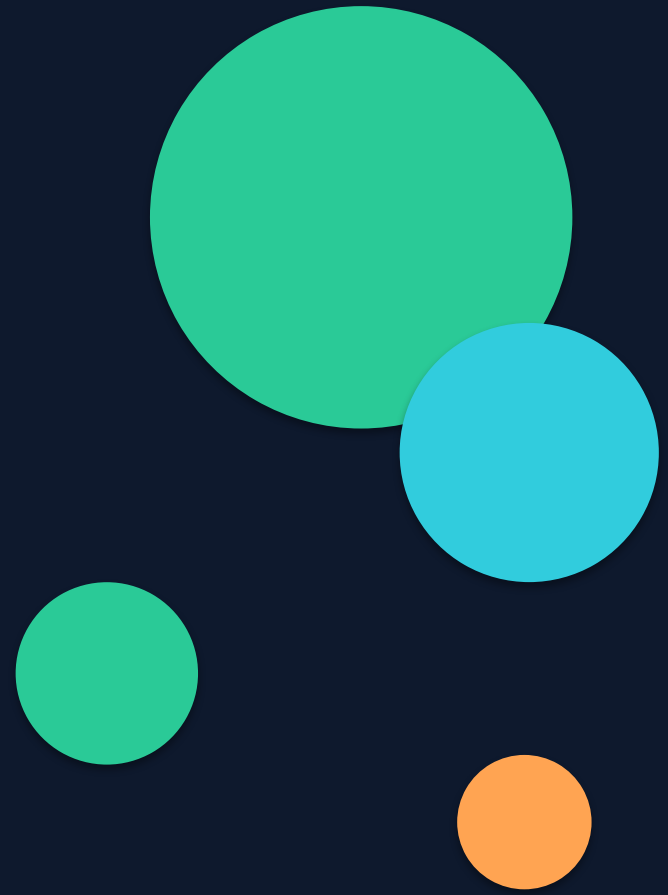
Exploratory Data Analysis & Visualization

Detailed compact notes with concepts, methods, examples and practical reminders.



6 topic slides + cover = 7 slides

Purpose and First Inspection

Key concepts • practical context • quick revision

- EDA summarizes data and helps discover structure, errors, unusual observations and candidate relationships.
- Inspect shape, column names, types, unique counts, missing values and duplicate records.
- Ask: what does one row represent, how was the data sampled, and what is the target question?
- Use head()/sample() for records, info() for types and describe() for numeric summaries.
- Check categorical frequencies and whether groups have very different sample sizes.
- EDA generates hypotheses; validate them rather than treating every pattern as a proven fact.

EXAMPLE / REMEMBER

Pandas: print(df.shape); print(df.info()); print(df.describe(include='all'))

Measures of Central Tendency and Spread

Key concepts • practical context • quick revision

- Mean = sum of values divided by count; it uses every value but is sensitive to extremes.
- Median is the middle sorted value and is often useful for skewed distributions.
- Mode is the most frequent value and works for categorical data too.
- Range = maximum − minimum; variance and standard deviation describe dispersion.
- Quartiles divide sorted data into four parts; IQR = Q3 − Q1.
- Compare mean and median, spread and sample size to understand a distribution.

EXAMPLE / REMEMBER

Example: incomes often have a high mean relative to median because a few values are very large.

Distribution, Skewness and Outliers

Key concepts • practical context • quick revision

- Histograms reveal shape, concentration, gaps, multiple peaks and skewness.
- Right skew has a long tail toward larger values; left skew has a tail toward smaller values.
- Box plots summarize median, quartiles and potential outliers.
- The 1.5 × IQR rule flags values beyond Q1 − 1.5×IQR or Q3 + 1.5×IQR.
- An outlier flag is a prompt for investigation, not proof that a record is wrong.
- Consider transformations or robust summaries when distributions are strongly skewed.

EXAMPLE / REMEMBER

Always inspect context before removing a point flagged as an outlier.

Choosing Visualizations

Key concepts • practical context • quick revision

- Bar chart: compare counts or values across categories; sort bars when helpful.
- Histogram: show a numeric distribution; adjust bins to reveal meaningful structure.
- Box/violin plot: compare distributions across groups.
- Scatter plot: inspect relationships between two numeric variables; add transparency for dense data.
- Line chart: show time trends; maintain chronological order and consistent intervals.
- Heatmap: compare a matrix, such as correlations, while labeling the scale clearly.

EXAMPLE / REMEMBER

Match the chart to the analytical question and data type.

Relationships and Group Comparisons

Key concepts • practical context • quick revision

- Scatter plots can reveal positive, negative, nonlinear or weak associations.
- Correlation summarizes linear association for numeric variables; it can miss nonlinear relationships.
- Correlation does not establish causation—confounding variables or reverse causality may explain a pattern.
- Group-by summaries compare means, medians, counts or rates across categories.
- Check sample size, uncertainty and potential confounders before interpreting group differences.
- Investigate surprising results and validate against source records and domain knowledge.

EXAMPLE / REMEMBER

A relationship between ice-cream sales and heat may be driven by temperature, not causation by sales.

Communicating EDA Findings

Key concepts • practical context • quick revision

- Use a direct title that states the insight, not just 'Chart 1'.
- Label axes, units, time periods and categories; use readable fonts and consistent scales.
- Avoid 3D effects, truncated axes that mislead, clutter and unnecessary decoration.
- Report sample size, missingness and key limitations alongside findings.
- Separate observed facts from interpretations and recommendations.
- Save code and record data versions so results can be reproduced.

EXAMPLE / REMEMBER

A good insight connects evidence to the question and clearly states uncertainty.